

DOCUMENTACIÓN DEL PROCESO

Instalación de Debian, Apache Hadoop y ejecución de MapReduce

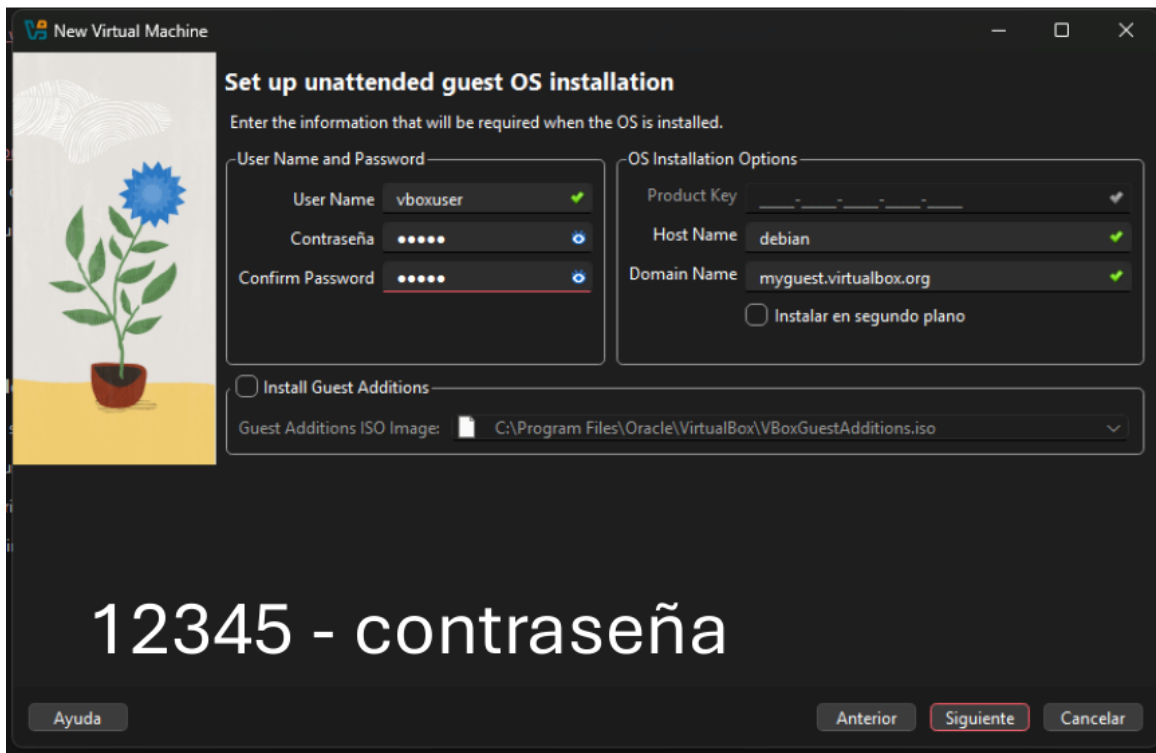
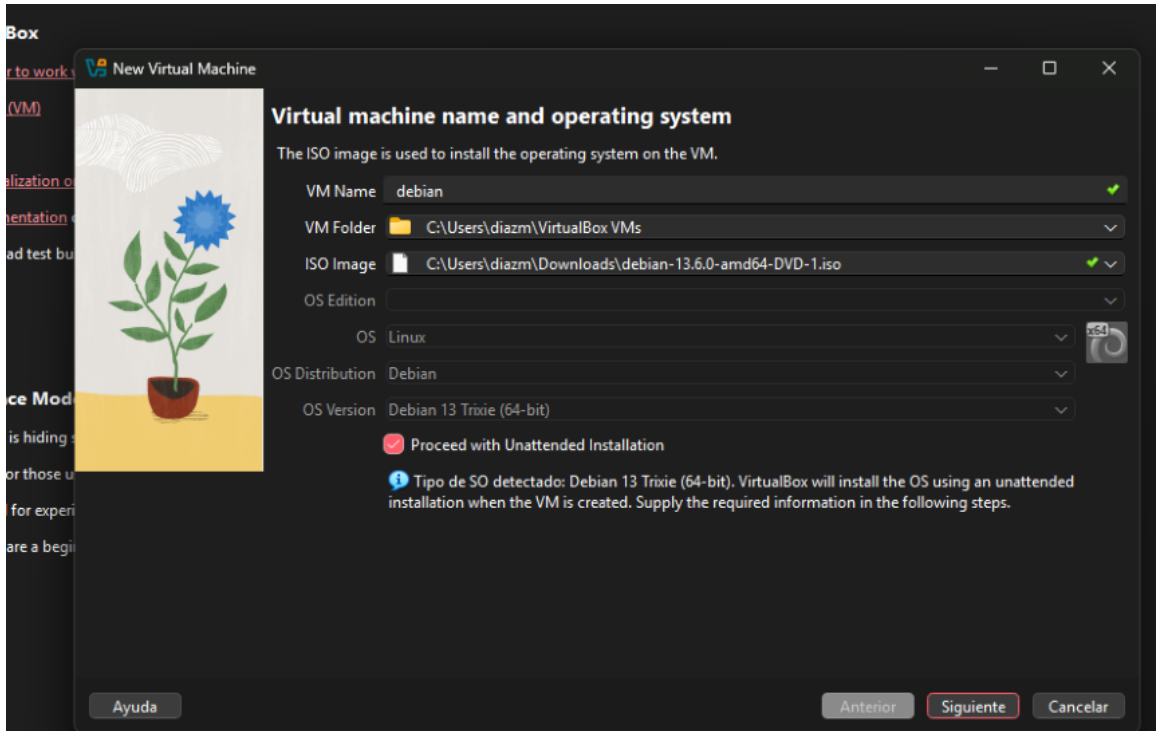
Guía breve basada en las evidencias capturadas durante la práctica



El presente documento resume el procedimiento realizado para crear una máquina virtual con Debian, instalar Apache Hadoop, configurar HDFS y YARN, y ejecutar un ejemplo de procesamiento distribuido mediante MapReduce. Cada apartado incluye la evidencia correspondiente y una explicación breve de su finalidad.

Presented by: *Lehidy Pamela Diaz Munayco*

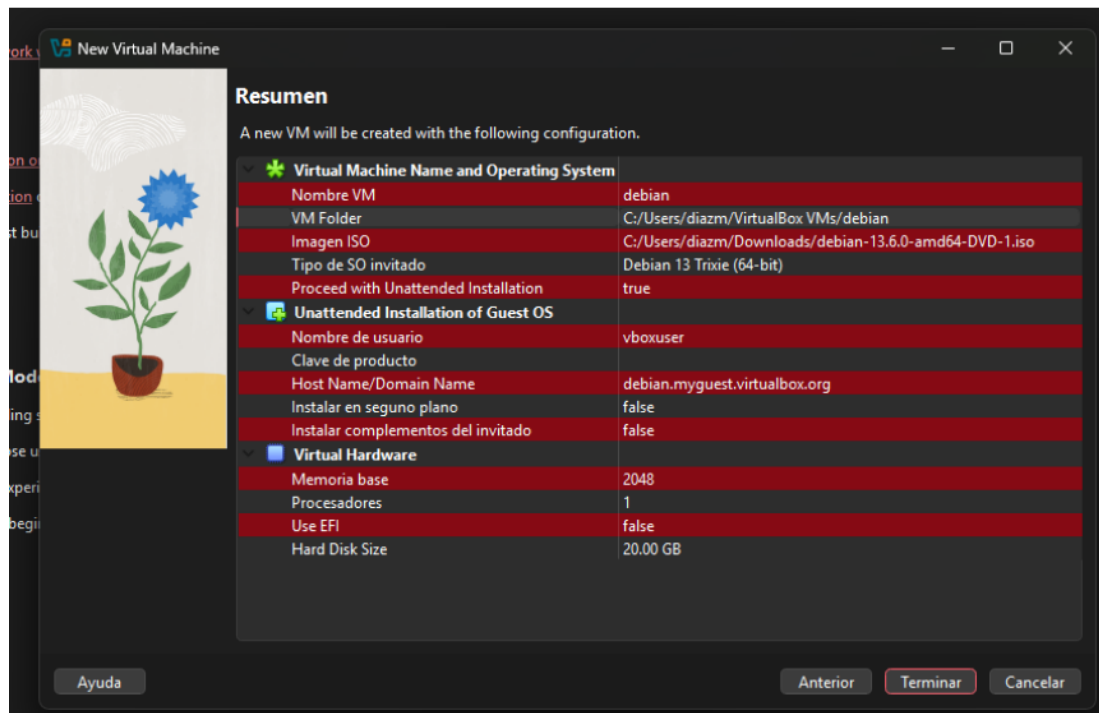
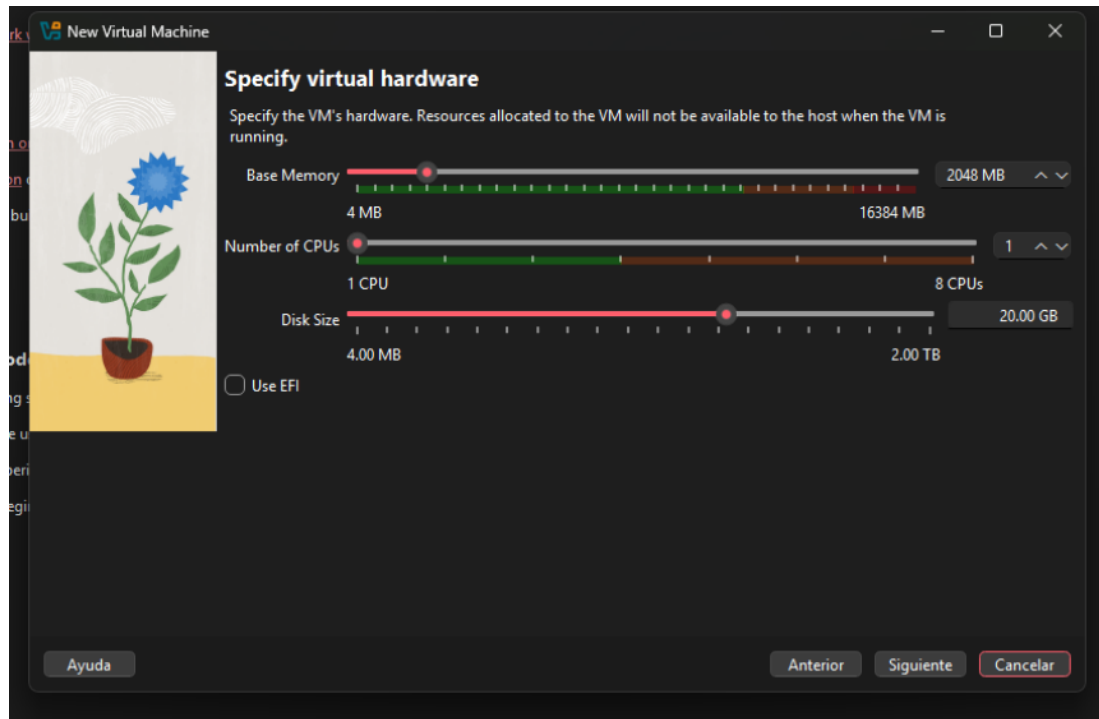
1. Creación de la máquina virtual



Evidencia del paso 1

Se creó una nueva máquina virtual en Oracle VirtualBox y se seleccionó la imagen ISO de Debian. También se definieron el nombre del usuario, el nombre del equipo y una contraseña para realizar la instalación automática del sistema operativo.

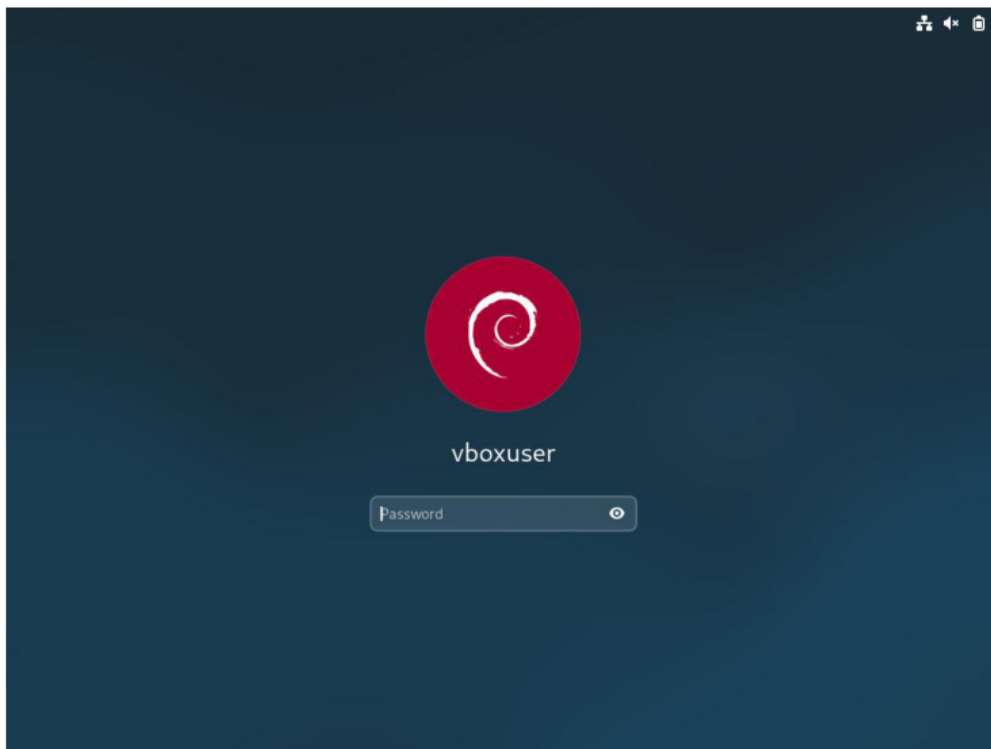
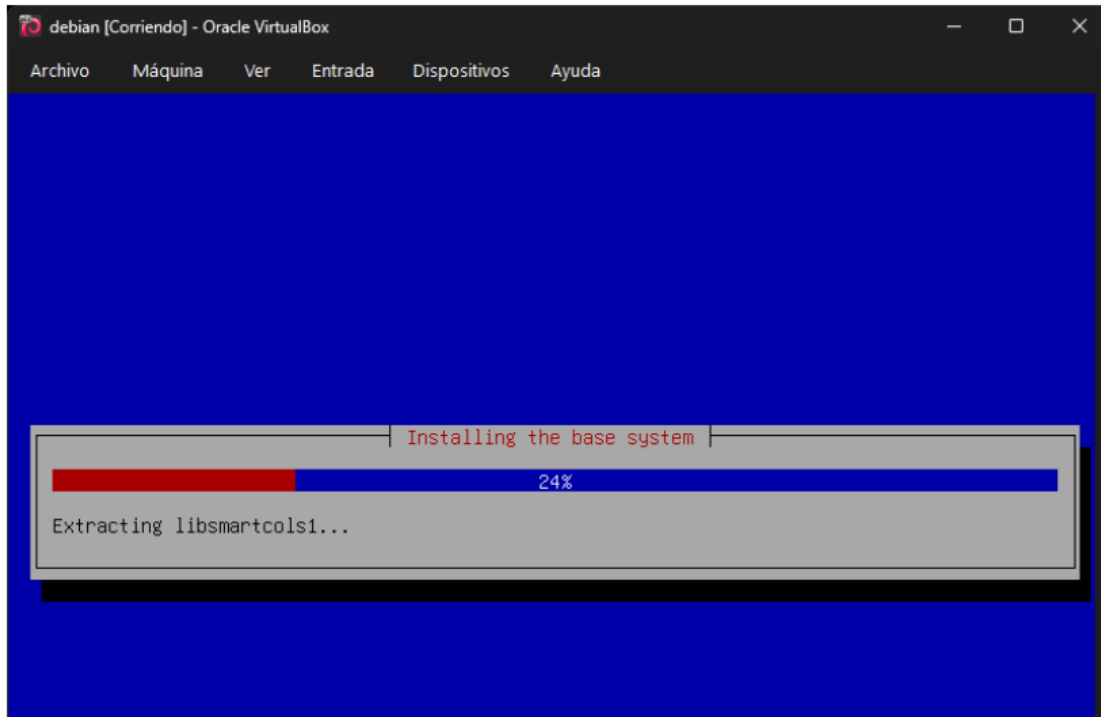
2. Asignación de recursos



Evidencia del paso 2

Se configuraron los recursos de la máquina virtual, asignando memoria RAM, procesador y espacio de almacenamiento. Estos recursos permiten que Debian y los servicios de Hadoop puedan ejecutarse dentro del entorno virtual.

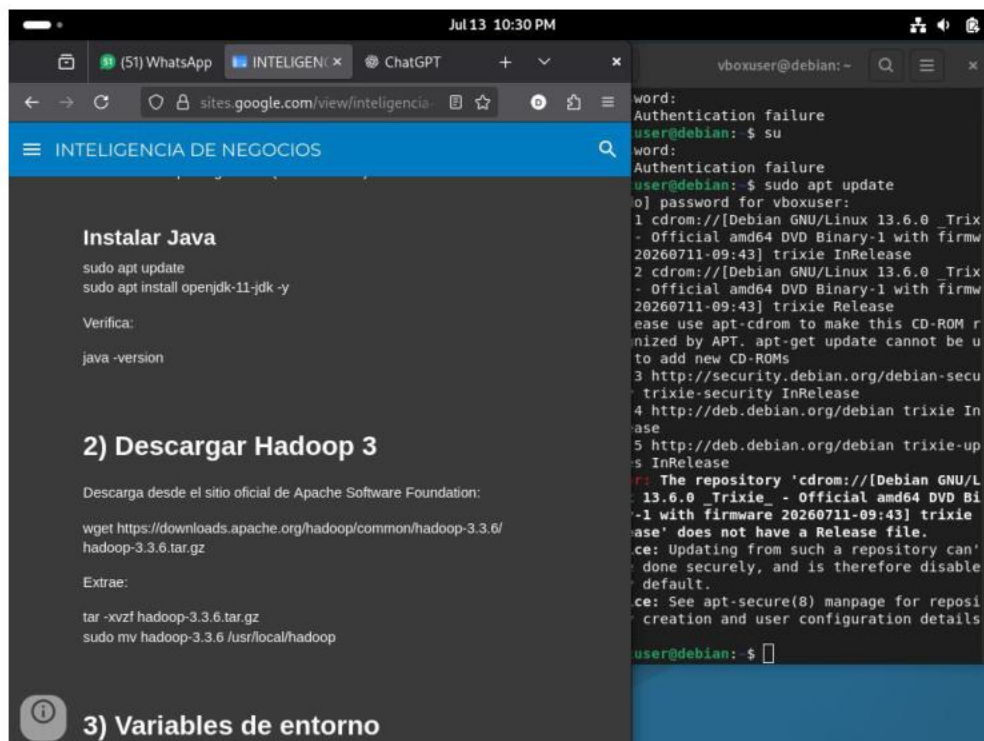
3. Instalación e inicio de Debian



Evidencia del paso 3

VirtualBox instaló el sistema base de Debian utilizando la imagen ISO seleccionada. Al finalizar, se accedió a la pantalla de inicio de sesión con el usuario creado durante la configuración.

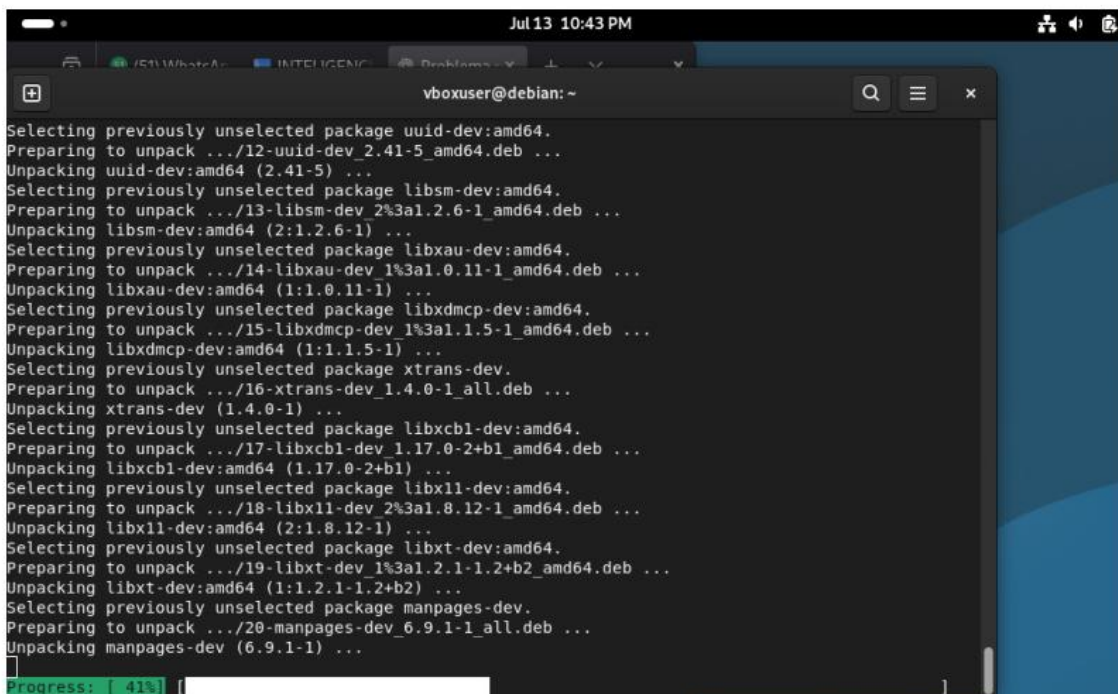
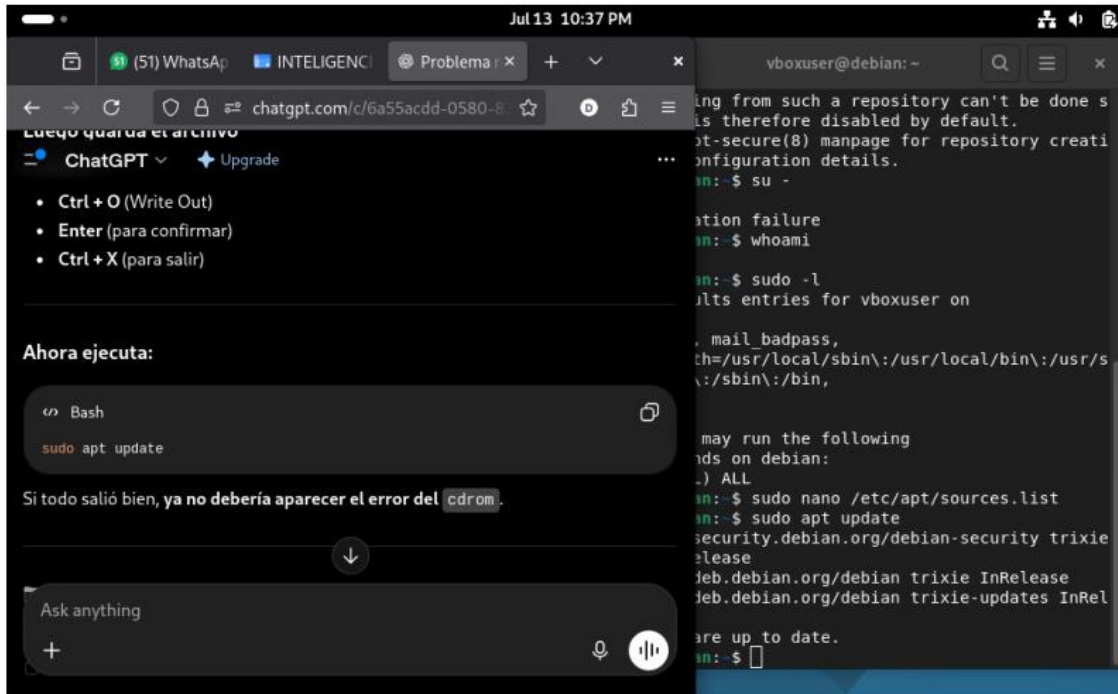
4. Primer inicio y actualización del sistema



Evidencia del paso 4

Después de ingresar a Debian, se abrió la terminal para actualizar los repositorios del sistema. En este paso se identificó un inconveniente con el repositorio del CD-ROM, el cual impedía completar correctamente la actualización.

5. Corrección de repositorios e instalación de paquetes



Evidencia del paso 5

Se editó el archivo de repositorios de Debian y se ejecutó nuevamente la actualización. Luego se instalaron los paquetes necesarios para preparar el sistema antes de configurar Java, SSH y Hadoop.

6. Instalación de Java y descarga de Hadoop

```
vboxuser@debian: ~
update-alternatives: using /usr/lib/jvm/java-25-openjdk-amd64/bin/jstat to provide /usr/bin/jstat (jstat) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-amd64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-amd64/bin/jwebserver to provide /usr/bin/jwebserver (jwebserver) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-amd64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-25-openjdk-amd64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up openjdk-25-jdk:amd64 (25.0.3+9-2-deb13u1) ...
update-alternatives: using /usr/lib/jvm/java-25-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
vboxuser@debian: $ java -version
openjdk version "25.0.3" 2026-04-21
OpenJDK Runtime Environment (build 25.0.3+9-2-deb13u1-Debian)
OpenJDK 64-Bit Server VM (build 25.0.3+9-2-deb13u1-Debian, mixed mode, sharing)
vboxuser@debian: $ javac -version
javac 25.0.3
vboxuser@debian: $
```



```
vboxuser@debian: ~
op-3.3.6.tar 80% 561.29M 8.11MB/s eta 18
p-3.3.6.tar. 80% 562.46M 8.01MB/s eta 18
-3.3.6.tar.g 80% 563.92M 8.01MB/s eta 18
3.3.6.tar.gz 81% 565.36M 7.73MB/s eta 18
.3.6.tar.gz 81% 566.54M 7.67MB/s eta 17
3.6.tar.gz 81% 568.15M 7.68MB/s eta 17
.6.tar.gz 81% 569.62M 7.34MB/s eta 17
6.tar.gz 82% 571.25M 7.38MB/s eta 17
.tar.gz 82% 572.98M 7.50MB/s eta 17
tar.gz 82% 574.60M 7.20MB/s eta 16
ar.gz 82% 576.19M 7.17MB/s eta 16
r.gz 82% 577.79M 7.21MB/s eta 16
.gz 83% 579.38M 7.16MB/s eta 16
gz 83% 579.54M 6.78MB/s eta 16
z 83% 580.08M 6.49MB/s eta 15
 83% 584.43M 7.50MB/s eta 15
    h 84% 585.96M 7.47MB/s eta 15
    ha 84% 587.84M 7.59MB/s eta 15
    had 84% 589.29M 7.66MB/s eta 15
    hado 84% 589.75M 7.36MB/s eta 14
    hadoo 85% 592.88M 7.84MB/s eta 14
    hadoop 85% 594.63M 7.97MB/s eta 14
    hadoop- 85% 596.31M 8.04MB/s eta 14
    hadoop-3 85% 597.99M 8.02MB/s eta 14
    hadoop-3. 86% 599.74M 8.18MB/s eta 12
    hadoop-3.3 86% 601.51M 8.21MB/s eta 12
    hadoop-3.3. 86% 603.25M 8.23MB/s eta 12
    hadoop-3.3.6 86% 604.99M 8.26MB/s eta 12
s
```

Evidencia del paso 6

Se instaló el entorno de desarrollo de Java y se verificó su funcionamiento mediante los comandos de versión. Posteriormente, se descargó Apache Hadoop 3.3.6 desde el repositorio oficial.

7. Ubicación de Hadoop y configuración de Java

```
vboxuser@debian: ~
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.2.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_2.9.2.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.0.0-alpha2.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.0.2.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_2.10.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.1.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.0.1.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.2.1.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.2.4.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/hadoop-h
dfs_0.21.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.1.3.xml
hadoop-3.3.6/share/hadoop/hdfs/hadoop-hdfs-cl
ient-3.3.6-tests.jar
hadoop-3.3.6/share/hadoop/hdfs/hadoop-hdfs-ht
tpfs-3.3.6.jar
vboxuser@debian:~$ sudo mv hadoop-3.3.6 /usr/
local/hadoop
vboxuser@debian:~$
```

```
vboxuser@debian: ~
adoop_HDFS_2.10.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.1.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.0.1.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.2.1.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.2.4.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/hadoop-h
dfs_0.21.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.1.3.xml
hadoop-3.3.6/share/hadoop/hdfs/hadoop-hdfs-cl
ient-3.3.6-tests.jar
hadoop-3.3.6/share/hadoop/hdfs/hadoop-hdfs-ht
tpfs-3.3.6.jar
vboxuser@debian:~$ sudo mv hadoop-3.3.6 /usr/
local/hadoop
vboxuser@debian:~$ ls /usr/local/
bin  games  include  libexec  sbin  src
etc  hadoop  lib      man      share
vboxuser@debian:~$ readlink -f $(which java)
/usr/lib/jvm/java-25-openjdk-amd64/bin/java
vboxuser@debian:~$ nano ~/.bashrc
vboxuser@debian:~$ source ~/.bashrc
vboxuser@debian:~$ echo $JAVA_HOME
/usr/lib/jvm/java-25-openjdk-amd64
vboxuser@debian:~$
```

Evidencia del paso 7

El directorio descargado de Hadoop se movió a /usr/local/hadoop para mantener una ubicación fija. Después se identificó la ruta de Java y se registró la variable JAVA_HOME dentro del archivo de configuración del usuario.

8. Variables de entorno y creación de claves SSH

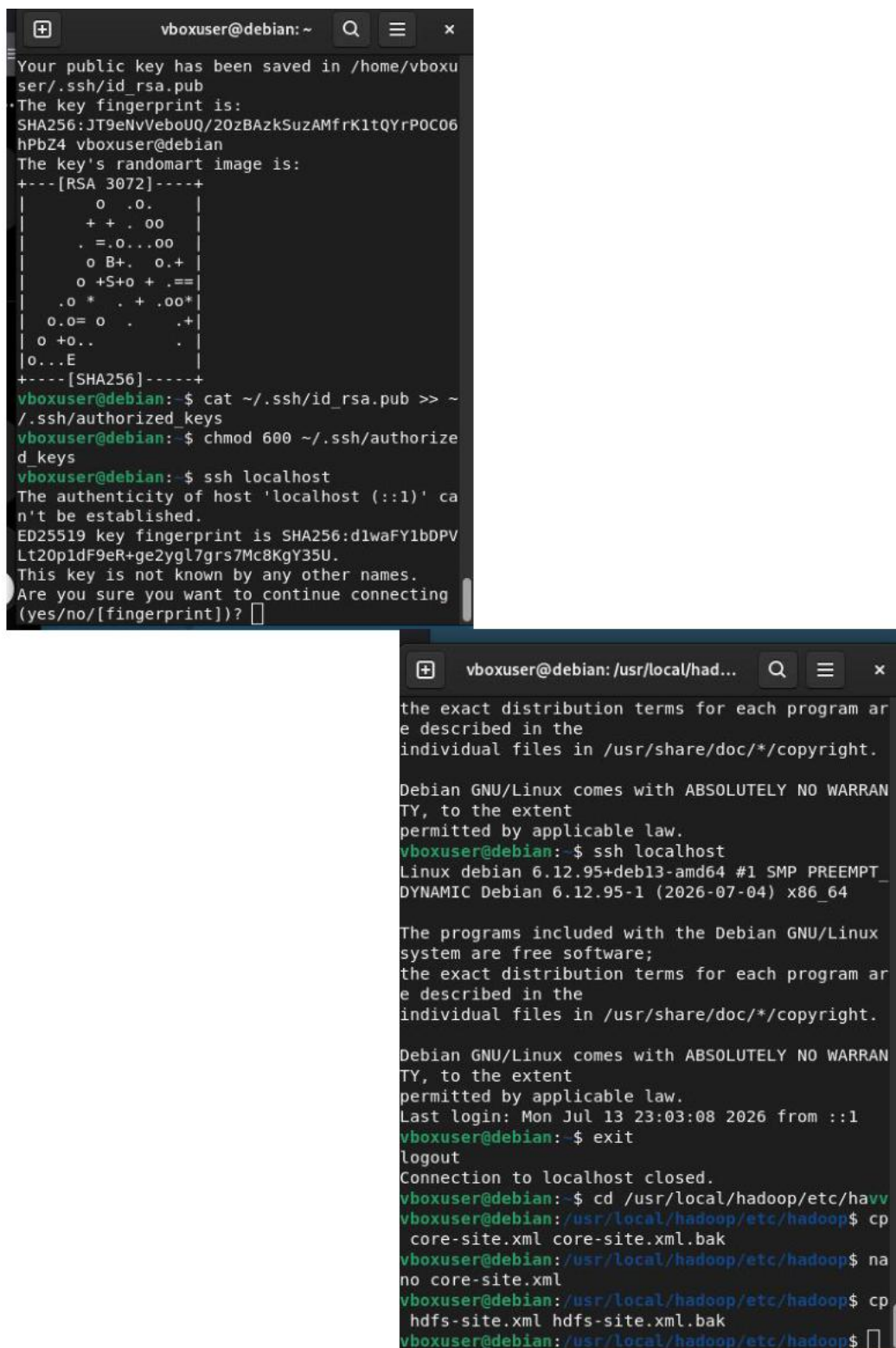
```
vboxuser@debian: ~
adoop_HDFS_3.1.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.0.1.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.2.1.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.2.4.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/hadoop-h
dfs_0.21.0.xml
hadoop-3.3.6/share/hadoop/hdfs/jdiff/Apache_H
adoop_HDFS_3.1.3.xml
hadoop-3.3.6/share/hadoop/hdfs/hadoop-hdfs-cl
ient-3.3.6-tests.jar
hadoop-3.3.6/share/hadoop/hdfs/hadoop-hdfs-ht
tpfs-3.3.6.jar
vboxuser@debian: ~$ sudo mv hadoop-3.3.6 /usr/
local/hadoop
vboxuser@debian: ~$ ls /usr/local/
bin  games  include  libexec  sbin  src
etc  hadoop  lib      man      share
vboxuser@debian: ~$ readlink -f $(which java)
/usr/lib/jvm/java-25-openjdk-amd64/bin/java
vboxuser@debian: ~$ nano ~/.bashrc
vboxuser@debian: ~$ source ~/.bashrc
vboxuser@debian: ~$ echo $JAVA_HOME
/usr/lib/jvm/java-25-openjdk-amd64
vboxuser@debian: ~$ echo $HADOOP_HOME
/usr/local/hadoop
vboxuser@debian: ~$
```

```
vboxuser@debian: ~
Created symlink '/etc/systemd/system/ssh.sock
et.wants/ssh-keygen.service' -> '/usr/lib/sys
temd/system/ssh-keygen.service'.
Processing triggers for man-db (2.13.1-1) ...
vboxuser@debian: ~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/vb
oxuser/.ssh/id_rsa):
Created directory '/home/vboxuser/.ssh'.
Your identification has been saved in /home/vb
oxuser/.ssh/id_rsa
Your public key has been saved in /home/vboxu
ser/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:JT9eNvVeboUQ/20zBAzkSuzAMfrK1tQYrPOC06
hPbZ4 vboxuser@debian
The key's randomart image is:
+---[RSA 3072]-----+
|
| o .o. |
| + + . oo |
| . = .o...oo |
| o B+. o.+ |
| o +S+o + .== |
| .o * . + .oo* |
| o.o= o . .+ |
| o +o.. . |
| o...E |
+---[SHA256]-----+
```

Evidencia del paso 8

Se configuró la variable HADOOP_HOME y se comprobó que apuntara al directorio correcto. Además, se generó una clave SSH, necesaria para que los servicios de Hadoop puedan comunicarse localmente sin solicitar la contraseña en cada conexión.

9. Acceso SSH sin contraseña y respaldo de archivos



```
vboxuser@debian: ~
Your public key has been saved in /home/vboxuser/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:JT9eNVVeboUQ/20zBAzkSuzAMfrK1tQYrPOC06hPbZ4
vboxuser@debian
The key's randomart image is:
+---[RSA 3072]-----+
|
| o .o. |
| + + . oo |
| . =.o...oo |
| o B+. o.+ |
| o +S+o + .==|
| .o * . + .oo+|
| o.o= o . .+|
| o +o.. . |
|o...E |
+---[SHA256]-----+
vboxuser@debian:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
vboxuser@debian:~$ chmod 600 ~/.ssh/authorized_keys
vboxuser@debian:~$ ssh localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is SHA256:dlwaFY1bDPV
Lt20p1dF9eR+ge2ygl7grs7Mc8KgY35U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? [y]

vboxuser@debian: /usr/local/hadoop...
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
vboxuser@debian:~$ ssh localhost
Linux debian 6.12.95+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.95-1 (2026-07-04) x86_64

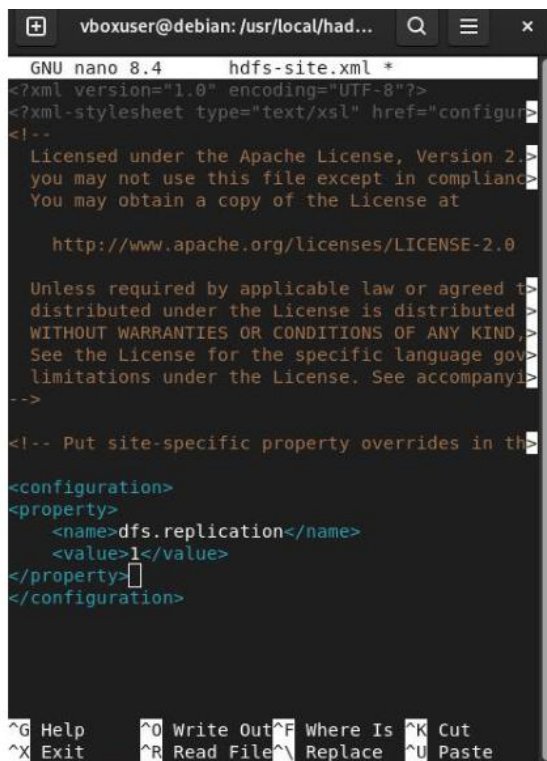
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jul 13 23:03:08 2026 from ::1
vboxuser@debian:~$ exit
logout
Connection to localhost closed.
vboxuser@debian:~$ cd /usr/local/hadoop/etc/hadoop
vboxuser@debian:/usr/local/hadoop/etc/hadoop$ cp core-site.xml core-site.xml.bak
vboxuser@debian:/usr/local/hadoop/etc/hadoop$ nano core-site.xml
vboxuser@debian:/usr/local/hadoop/etc/hadoop$ cp hdfs-site.xml hdfs-site.xml.bak
vboxuser@debian:/usr/local/hadoop/etc/hadoop$
```

Evidencia del paso 9

La clave pública se agregó al archivo `authorized_keys` y se probó la conexión mediante `ssh localhost`. Luego se realizaron copias de seguridad de los archivos principales antes de modificar la configuración de Hadoop.

10. Configuración de HDFS, MapReduce y YARN



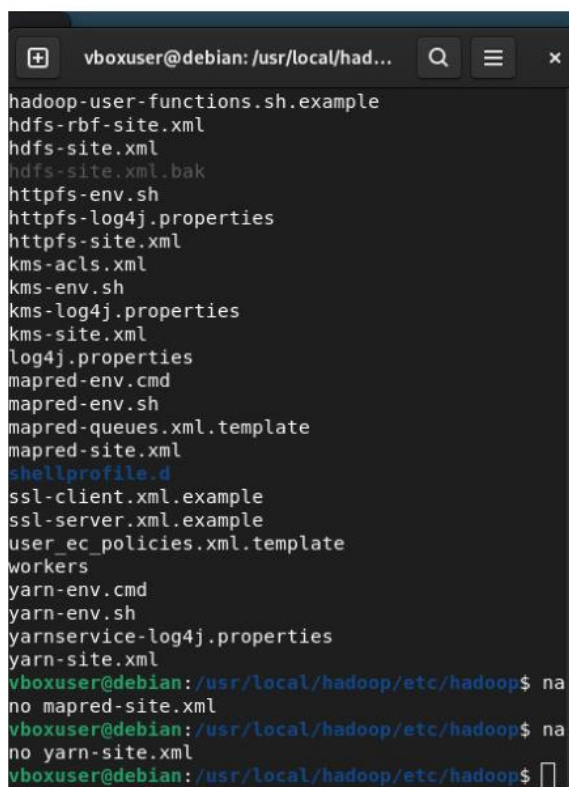
```

GNU nano 8.4 hdfs-site.xml *
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl" />
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at
    http://www.apache.org/licenses/LICENSE-2.0
  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file for details.
-->

<!-- Put site-specific property overrides in this file -->

<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
</configuration>

```



```

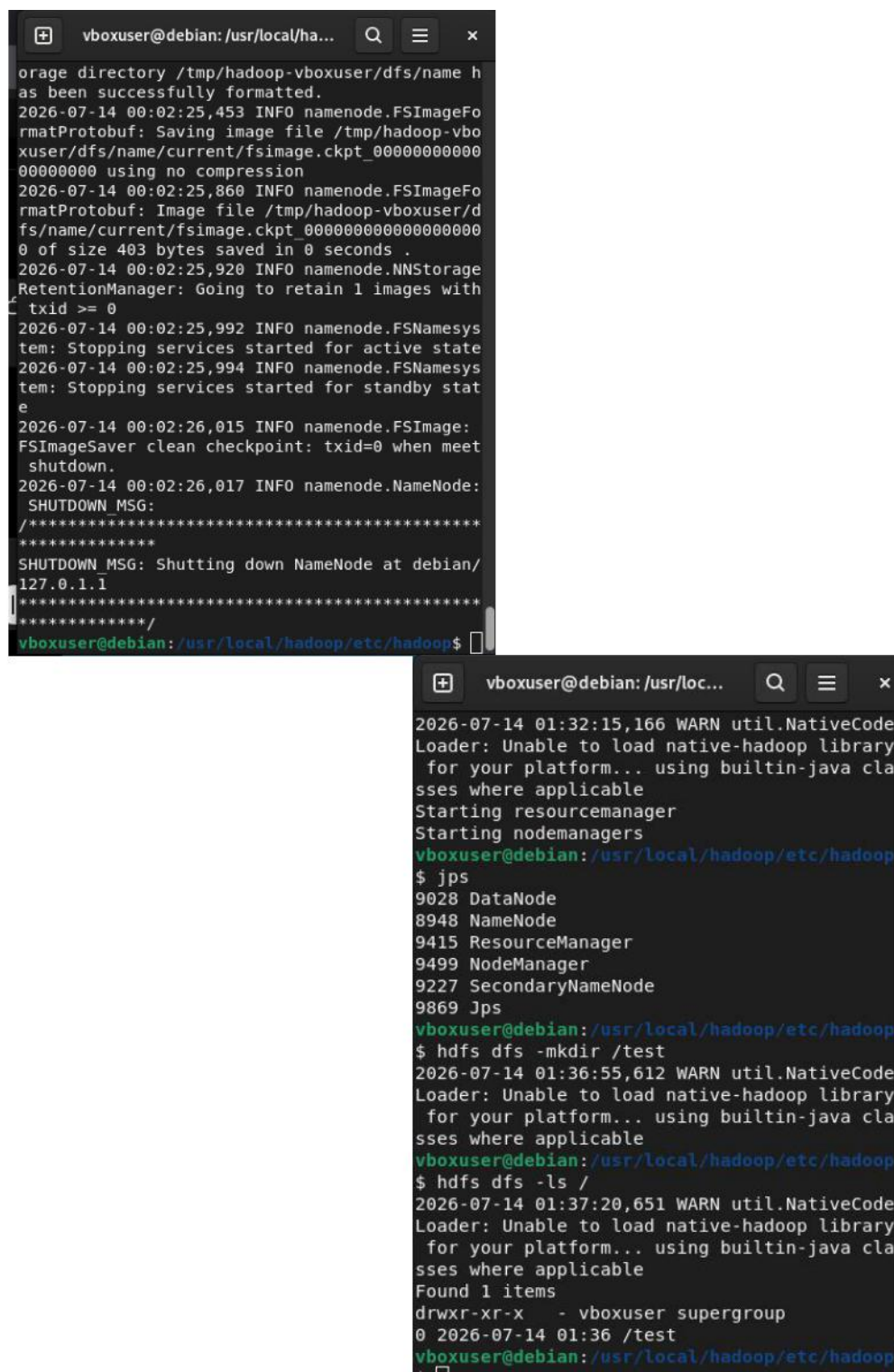
vboxuser@debian: /usr/local/hadoop...
hadoop-user-functions.sh.example
hdfs-rbf-site.xml
hdfs-site.xml
hdfs-site.xml.bak
https-env.sh
https-log4j.properties
https-site.xml
kms-acls.xml
kms-env.sh
kms-log4j.properties
kms-site.xml
log4j.properties
mapred-env.cmd
mapred-env.sh
mapred-queues.xml.template
mapred-site.xml
shellprofile.d
ssl-client.xml.example
ssl-server.xml.example
user_ec_policies.xml.template
workers
yarn-env.cmd
yarn-env.sh
yarnservice-log4j.properties
yarn-site.xml
vboxuser@debian: /usr/local/hadoop/etc/hadoop$ na
no mapred-site.xml
vboxuser@debian: /usr/local/hadoop/etc/hadoop$ na
no yarn-site.xml
vboxuser@debian: /usr/local/hadoop/etc/hadoop$

```

Evidencia del paso 10

Se editaron los archivos hdfs-site.xml, mapred-site.xml y yarn-site.xml. Estas configuraciones establecen la replicación de HDFS, indican que MapReduce trabajará sobre YARN y habilitan el servicio de intercambio de datos entre las tareas.

11. Formateo de HDFS e inicio de servicios



The image consists of two terminal window screenshots. The top window shows the successful formatting of the HDFS NameNode and its subsequent shutdown. The bottom window shows the startup of Hadoop services (NameNode, DataNode, ResourceManager, NodeManager, and SecondaryNameNode) and the creation of a test directory in HDFS.

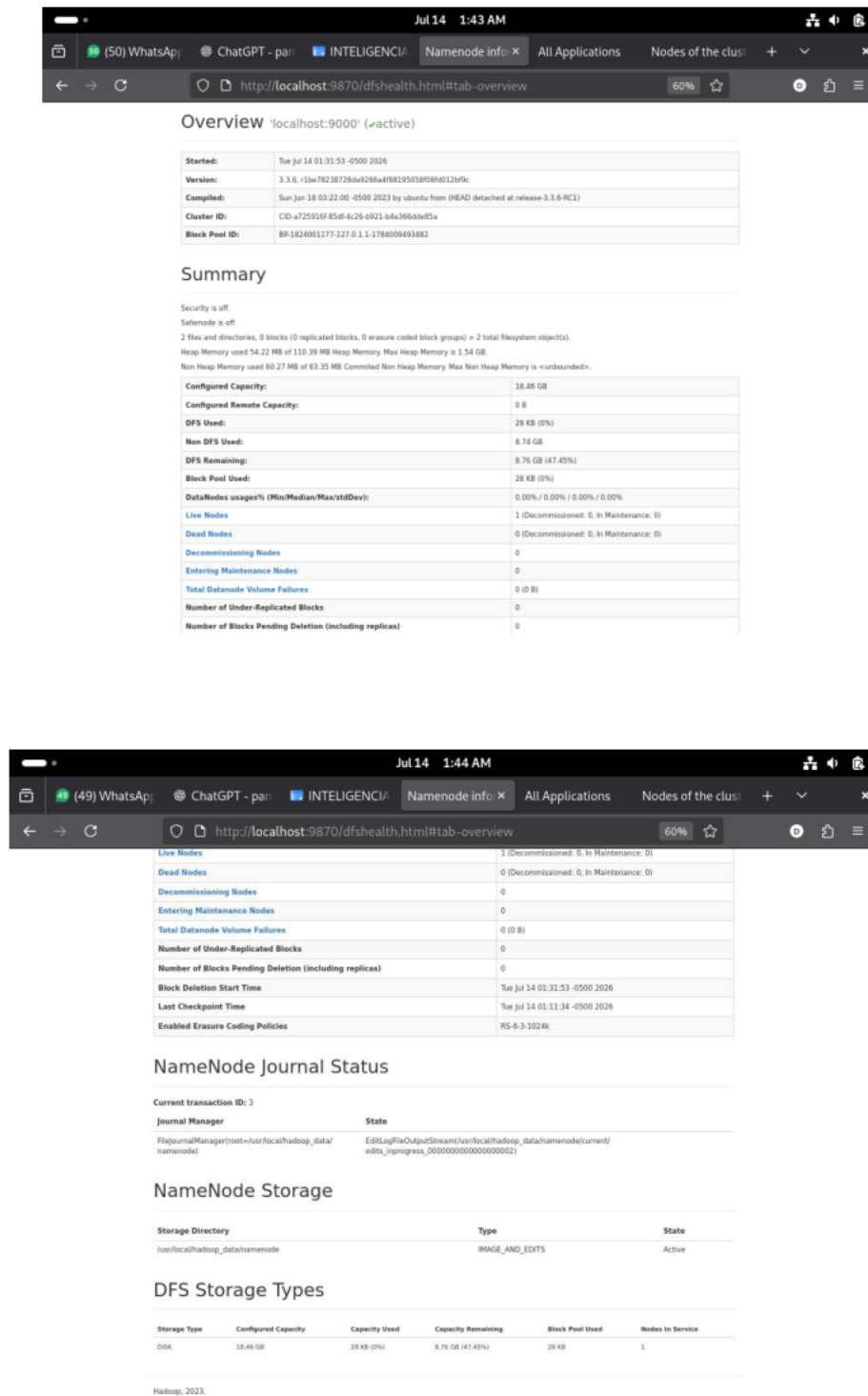
```
vboxuser@debian: /usr/local/ha...
orage directory /tmp/hadoop-vboxuser/dfs/name h
as been successfully formatted.
2026-07-14 00:02:25,453 INFO namenode.FSImageFo
rmatProtobuf: Saving image file /tmp/hadoop-vbo
xuser/dfs/name/current/fsimage.ckpt_0000000000
00000000 using no compression
2026-07-14 00:02:25,860 INFO namenode.FSImageFo
rmatProtobuf: Image file /tmp/hadoop-vboxuser/d
fs/name/current/fsimage.ckpt_00000000000000000
0 of size 403 bytes saved in 0 seconds .
2026-07-14 00:02:25,920 INFO namenode.NNStorage
RetentionManager: Going to retain 1 images with
txid >= 0
2026-07-14 00:02:25,992 INFO namenode.FSNamesys
tem: Stopping services started for active state
2026-07-14 00:02:25,994 INFO namenode.FSNamesys
tem: Stopping services started for standby stat
e
2026-07-14 00:02:26,015 INFO namenode.FSImage:
FSImageSaver clean checkpoint: txid=0 when meet
shutdown.
2026-07-14 00:02:26,017 INFO namenode.NameNode:
SHUTDOWN MSG:
/*****
*****/
SHUTDOWN MSG: Shutting down NameNode at debian/
127.0.1.1
*****/
vboxuser@debian: /usr/local/hadoop/etc/hadoop$

vboxuser@debian: /usr/loc...
2026-07-14 01:32:15,166 WARN util.NativeCode
Loader: Unable to load native-hadoop library
for your platform... using builtin-java cla
sses where applicable
Starting resourcemanager
Starting nodemanagers
vboxuser@debian: /usr/local/hadoop/etc/hadoop
$ jps
9028 DataNode
8948 NameNode
9415 ResourceManager
9499 NodeManager
9227 SecondaryNameNode
9869 Jps
vboxuser@debian: /usr/local/hadoop/etc/hadoop
$ hdfs dfs -mkdir /test
2026-07-14 01:36:55,612 WARN util.NativeCode
Loader: Unable to load native-hadoop library
for your platform... using builtin-java cla
sses where applicable
vboxuser@debian: /usr/local/hadoop/etc/hadoop
$ hdfs dfs -ls /
2026-07-14 01:37:20,651 WARN util.NativeCode
Loader: Unable to load native-hadoop library
for your platform... using builtin-java cla
sses where applicable
Found 1 items
drwxr-xr-x - vboxuser supergroup
0 2026-07-14 01:36 /test
vboxuser@debian: /usr/local/hadoop/etc/hadoop
```

Evidencia del paso 11

Se formateó el NameNode para inicializar el sistema de archivos distribuido. Después se iniciaron HDFS y YARN, y con el comando `jps` se verificó la ejecución de NameNode, DataNode, SecondaryNameNode, ResourceManager y NodeManager.

12. Verificación del NameNode



The screenshot shows the Hadoop NameNode web interface in a browser. The address bar displays `http://localhost:9870/dfshealth.html#tab-overview`. The interface is divided into several sections:

Overview 'localhost:9000' (✓active)

Started:	Tue Jul 14 01:31:53 -0500 2026
Version:	3.3.6, r12w78238728d6d8a4f8819505808d0121f9c
Compiled:	Sun Jun 18 03:22:00 -0500 2023 by ubuntu from (HEAD detached at release-3.3.6-RC1)
Cluster ID:	CID-a725916f-85d4-4c26-b021-b4c366a6d85a
Block Pool ID:	BP-1824001177-127.0.1.1-1784000493482

Summary

Security is off.
Safemode is off.
2 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 2 total filesystem object(s).
Heap Memory used 54.22 MB of 110.39 MB Heap Memory. Max Heap Memory is 1.54 GB.
Non Heap Memory used 60.27 MB of 63.35 MB Committed Non Heap Memory. Max Non Heap Memory is «unbounded».

Configured Capacity:	18.48 GB
Configured Remote Capacity:	0 B
DFS Used:	28 KB (0%)
Non DFS Used:	8.74 GB
DFS Remaining:	8.76 GB (47.45%)
Block Pool Used:	28 KB (0%)
DataNodes usage% (Min/Median/Max/stdDev):	0.00% / 0.00% / 0.00% / 0.00%
Live Nodes	1 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes	0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes	0
Entering Maintenance Nodes	0
Total Datanode Volume Failures	0 (0 B)
Number of Under-Replicated Blocks	0
Number of Blocks Pending Deletion (including replicas)	0

NameNode Journal Status

Current transaction ID: 3

Journal Manager	State
FileJournalManager(root=/usr/local/hadoop_data/namenode)	EditingFileOutputStream(/usr/local/hadoop_data/namenode/current/edits_inprogress_00000000000000000002)

NameNode Storage

Storage Directory	Type	State
/usr/local/hadoop_data/namenode	IMAGE_AND_EDITS	Active

DFS Storage Types

Storage Type	Configured Capacity	Capacity Used	Capacity Remaining	Block Pool Used	Nodes In Service
Disk	18.48 GB	28 KB (0%)	8.76 GB (47.45%)	28 KB	1

Hadoop, 2023.

Evidencia del paso 12

Se ingresó a la interfaz web del NameNode mediante `localhost:9870`. Desde esta página se revisó el estado de HDFS, la capacidad de almacenamiento, los nodos activos y la información del directorio utilizado por el NameNode.

13. Verificación de YARN

All Applications

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used Resources	Total Resources
0	0	0	0	0	<memory:0 B, vCores:0>	<memory:8 GB, vCores:8>

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes
1	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation
Capacity Scheduler	[memory-mb (unit=MiB), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>

Show 20 entries

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU V-Cores	Allocated Memory MB	Allocated V-Cores
No data available in table															

Showing 0 to 0 of 0 entries.

Nodes of the cluster

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used Resources	Total Resources	Reserved Resources	Physical Mem Used %	Physical V-Cores Used %
0	0	0	0	0	<memory:0 B, vCores:0>	<memory:8 GB, vCores:8>	0B	88	82

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes	Shutdown Nodes
1	0	0	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation	Maximum Cluster Application Priority	Scheduler Busy %
Capacity Scheduler	[memory-mb (unit=MiB), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>	0	0

Show 20 entries

Node Labels	Rack	Node State	Node Address	Node HTTP Address	Last health-report	Health-report	Containers	Allocation Tags	Mem Used	Mem Avail	Phys Mem Used %	V-Cores Used	V-Cores Avail	Phys V-Cores Used %	Version
/ default rack	RUNNING	debian.org/debian-virtualbox.org:35143	debian.org/debian-virtualbox.org:8042	Jul Jul 14 01:42:35 -0500 2028	0	0	0	0	0	0	0	0	8	66	3.3.6

Showing 1 to 1 of 1 entries

First Previous 1 Next Last

Evidencia del paso 13

Se abrió la interfaz de YARN en localhost:8088. En ella se comprobó el funcionamiento del ResourceManager, el estado del nodo del clúster y el espacio disponible para ejecutar aplicaciones de procesamiento.

14. Ejecución del ejemplo WordCount

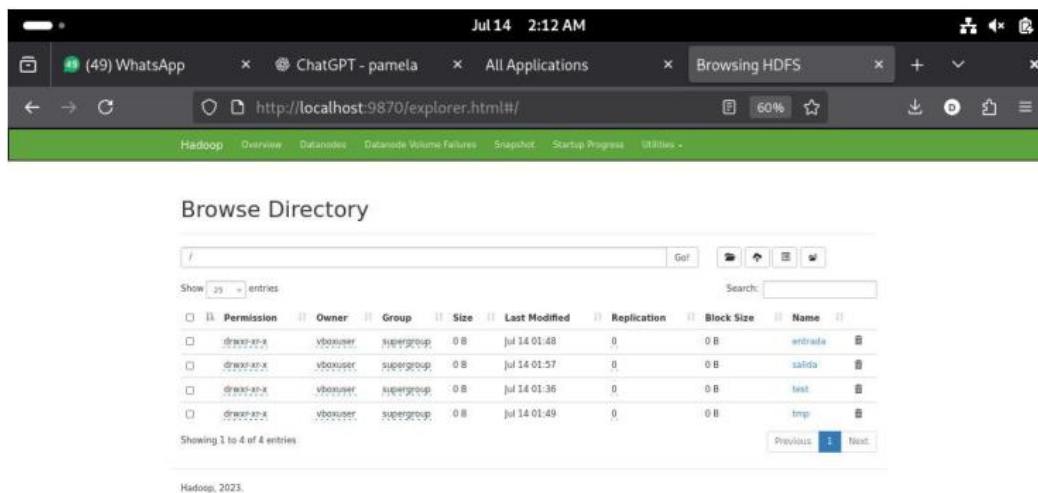
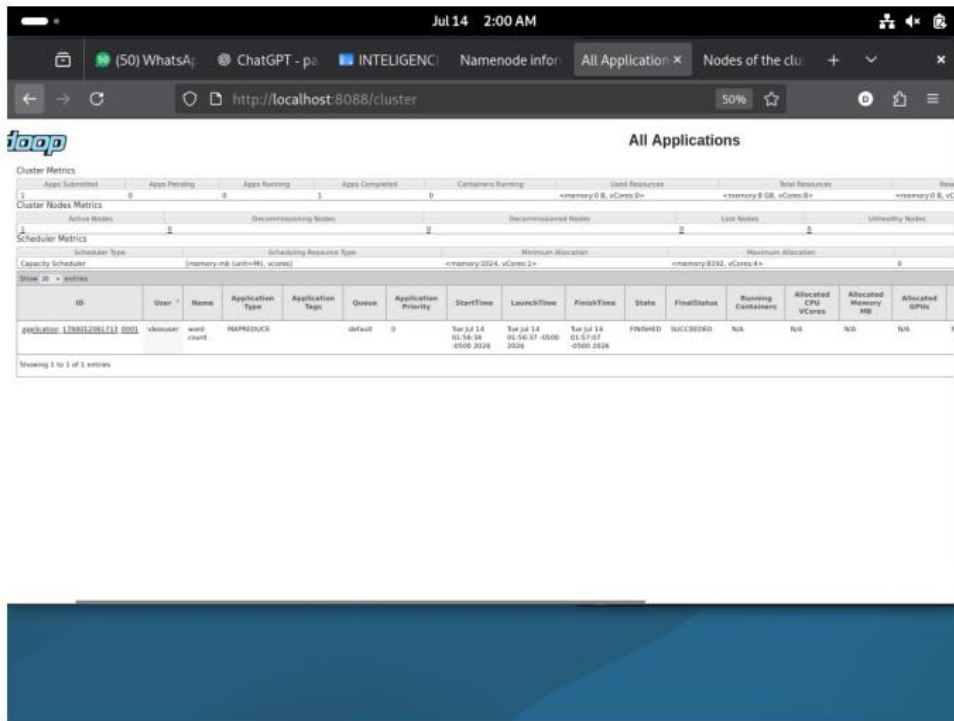
```
vboxuser@debian: /usr/loc...
0 2026-07-14 01:57 /salida/ SUCCESS
-rw-r--r-- 1 vboxuser supergroup 17
9 2026-07-14 01:57 /salida/part-r-00000
vboxuser@debian: /usr/local/hadoop/etc/hadoop
$ hdfs dfs -cat /salida/part-r-00000
2026-07-14 01:58:35,965 WARN util.NativeCode
Loader: Unable to load native-hadoop library
for your platform... using builtin-java cla
sses where applicable
big 1
cantidades 1
con 1
data 1
datos 1
de 2
distribuido 1
es 1
grandes 1
hadoop 3
hdfs 1
herramienta 1
mapreduce 1
permite 1
procesa 1
procesamiento 1
trabaja 1
una 1
y 1
yarn 1
vboxuser@debian: /usr/local/hadoop/etc/hadoop
```

```
vboxuser@debian: /usr/loc...
rm: `/salida': No such file or directory
vboxuser@debian: /usr/local/hadoop/etc/hadoop
$ jps
12368 SecondaryNameNode
12882 ResourceManager
12967 NodeManager
12071 NameNode
12154 DataNode
13419 Jps
vboxuser@debian: /usr/local/hadoop/etc/hadoop
$ hadoop jar $HADOOP_HOME/share/hadoop/mapre
duce/hadoop-mapreduce-examples-3.3.6.jar wor
dcount /entrada /salida
2026-07-14 01:56:27,515 WARN util.NativeCode
Loader: Unable to load native-hadoop library
for your platform... using builtin-java cla
sses where applicable
2026-07-14 01:56:29,891 INFO client.DefaultN
oHARMFailoverProxyProvider: Connecting to Re
sourceManager at /0.0.0.0:8032
2026-07-14 01:56:31,152 INFO mapreduce.JobRe
sourceUploader: Disabling Erasure Coding for
path: /tmp/hadoop-yarn/staging/vboxuser/.st
aging/job_1784012061713_0001
2026-07-14 01:56:32,585 INFO input.FileInput
Format: Total input files to process : 1
2026-07-14 01:56:32,834 INFO mapreduce.JobSu
bmitter: number of splits:1
2026-07-14 01:56:33,540 INFO mapreduce.JobSu
bmitter: Submitting tokens for job: job_1784
012061713_0001
```

Evidencia del paso 14

Se ejecutó el ejemplo WordCount incluido en Hadoop, utilizando un archivo almacenado en el directorio /entrada de HDFS. MapReduce procesó el contenido y generó los resultados dentro del directorio /salida.

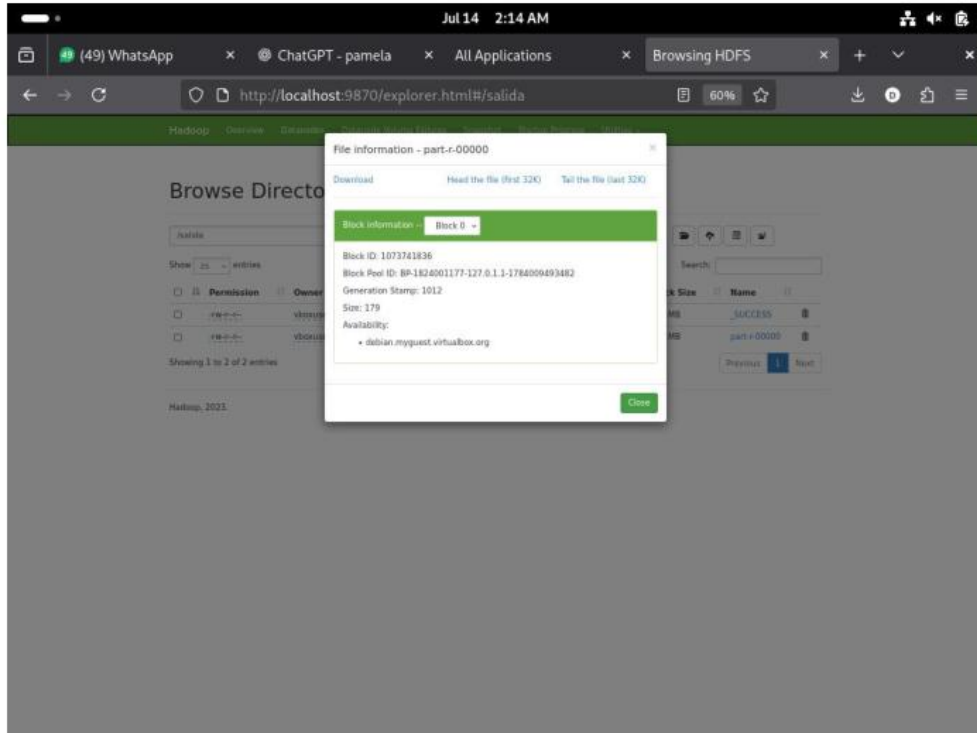
15. Comprobación del trabajo realizado



Evidencia del paso 15

La interfaz de YARN mostró la aplicación WordCount con estado FINISHED y resultado SUCCEEDED. Asimismo, el explorador de HDFS permitió observar los directorios de entrada, salida y prueba creados durante la actividad.

16. Revisión del archivo de salida



Evidencia del paso 16

Finalmente, se ingresó al directorio /salida y se revisó el archivo part-r-00000. Este archivo contiene el conteo de palabras producido por MapReduce y confirma que el proceso fue almacenado correctamente en HDFS.

Conclusión

La práctica permitió implementar un entorno Hadoop en modo pseudodistribuido dentro de Debian. La correcta ejecución de los servicios de HDFS y YARN, junto con el resultado exitoso de WordCount, demuestra que el sistema quedó preparado para almacenar archivos y ejecutar tareas básicas de procesamiento de datos mediante MapReduce.

Link del drive para descargar Debian

<https://drive.google.com/drive/folders/1boa3U9ohOFShDYZNtnGgrjcAun8O6YJU?usp=sharing>